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Hip joint

Course notes intended for first-year medical students.

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Hip joint

Course outline

Introduction
1. Articular surfaces
1.1. Acetabulum
1.2. Acetabular Labrum
1.3. Femoral head
2. Means of union
2.1. Joint capsule
2.2. Passive ligaments
2.3. Ligament of the femoral head
2.4. Active ligaments (the periarticular muscles)
3. Synovial
4. Joint physiology
5. Clinical applications
Conclusion
Bibliography

Hip joint

Introduction

The hip joint, or coxofemoral joint, is the single joint that connects the lower limb to the trunk. It is a proximal ball-and-socket joint of the lower limb with a dual function: support and stability.
The hip supports the pelvis and orients the lower limb in all directions of space, ensuring the upright posture that characterizes humans. It is a deep, weight-bearing, mobile, and highly congruent joint.

1.Articular surfaces

1.1.The acetabulum

The acetabulum is a deep cavity located on the lateral surface of the hip bone, bordered by a prominent bony rim called the acetabular limbus. The acetabulum consists of two distinct parts:

- A central, non-articular part: the acetabular fossa.
- A peripheral, articular, crescent-shaped part: the semilunar surface.

1.2.The acetabular labrum

The acetabular labrum is a fibrocartilaginous ring attached the perimeter of the acetabulum; it passes over the ischiopubic notch, forming the transverse ligament of the acetabulum.
It equalizes the acetabular limbus and increases the depth and extent of the acetabulum.

1.3.The femoral head

The femoral head is a rounded articular projection, representing 2/3 of a sphere. At its center is the fovea capitis where the ligament of the femoral head (the ligamentum teres) attaches.

2. Means of union

The following maintain the joint surfaces in contact: the joint capsule, the passive ligaments, the ligament of the femoral head and the passive ligaments.

2.1.The joint capsule

The joint capsule is a fibrous sleeve which has an iliac insertion on the acetabular limbus and labrum and a femoral insertion on the intertrochanteric line and femoral neck.

2.2.Passive ligaments

2.2.1. Iliofemoral ligament (Bertin ligament)

The anterior ligament, triangular and strong, extends between the acetabulum and the intertrochanteric line. It has two distinct bundles :

- the superior bundle : ilio-pretrochanteric
- the inferior ilio-pretrochanteric bundle.

2.2.2. pubofemoral ligament

The pubofemoral ligament reinforces the anteroinferior part of the joint capsule and extends between the iliopectineal eminence and the pretrochanteric depression.

2.2.3. Ischiofemoral ligament

The ischiofemoral ligament reinforces the posterior joint capsule and extends between the ischium and femoral neck.

2.3. Ligament of the femoral head (round ligament)

The ligament of the femoral head is an intra-articular and extra-synovial ligament, centered on an artery that supplies the femoral head. It extends from the fovea capitis and terminates in three bundles:

- Anterior bundle: on the anterior horn of the semilunar surface.
- The middle bundle attaches to the transverse ligament.
- Posterior bundle on the ischium.

2.4.Active ligaments (periarticular muscles)

- Anterior : the Iliopsoas muscle, the rectus femoris muscle and the sartorius muscle.
- Lateral : the tensor fasciae latae muscle, the gluteus medius muscle, and the gluteus minimus muscle.
- Posterior : the gluteus maximus muscle, the pelvi-trochanteric muscles and the hamstring muscles.
- Medial : the pectineus muscle, the adductor longus muscle, the adductor magnus muscle

3. Synovial

The synovium lines the deep surface of the capsule.

The synovium of the ligament of the femoral head is separate and independent from the articular synovium; it attaches to the periphery of the fovea capitis and the acetabular fossa.

4. Joint physiology

The hip joint is a very stable joint with a wide range of motion. Its stability is ensured by the joint capsule, ligaments, and surrounding muscles. It allows:

- Flexion-extension movements
- Adduction-abduction movements
- Medial rotation – lateral rotation movements
- The circumduction

4.1.Flexion-extension movements

Flexion and extension occur around a transverse axis that passes through the center of the femoral head.

Flexion brings the anterior surface of the thigh closer to the anterior abdominal wall.
The range of motion: 120°: knee bent, 90°: knee extended.
The extension moves the anterior surface of the thigh away from the anterior abdominal wall
The range of motion: 10° with the knee bent and 20° with the knee extended

4.2.Adduction-abduction movements
Abduction and adduction occur around an anteroposterior axis that passes through the center of the femoral head.
Abduction spread the thigh away from the midline
The usual range of motion: 30°, forced: 45° to 60° (gymnastics…)
Adduction brings the thigh closer to the midline.
Maximum angle of rotation : 30°

4.3.The circumduction motion
Circumduction results from the succession of the 4 preceding movements.

4.4.The medial rotation – lateral rotation movements performed around a vertical axis that passes through the center of the femoral head, lateral rotation: 45°- 60°, medial rotation: 30°- 45°.

5. Clinical application

5.1.Radioanatomy
Numerous techniques are available today for analyzing the tendons, bone, capsuloligamentous structures, fibrocartilage, trophic state, and muscular content of the hip joint. These techniques utilize ultrasound, computed tomography (CT) scans, with or without arthrography, and magnetic resonance imaging (MRI), with or without arthrography.

5.2.Pathologies

- Osteoarthritis of the hip joint or coxarthrosis.
- Arthritis is a serious inflammation or infection (it can be treated with prostheses).
- Fractures of the femoral neck or acetabulum.
- Dislocations are rare because of the stability of the joint.
- Congenital hip dislocations (CHD) are detected at birth using the Ortolani maneuver (which assesses instability during hip mobilization). Orthopedic treatment involves placing the baby in a swaddling position with their hips in forced abduction.

Conclusion

The hip joint is a weight-bearing, ball-and-socket joint connecting the pelvis to the femur. It is both highly mobile and extremely stable thanks to its congruent structure and powerful connecting elements such as the capsule, ligaments, and muscles. Its anatomy allows for a wide range of motion essential for walking and posture, while simultaneously resolving the paradox of stability, making it one of the strongest joints in the human body.

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